On considère la suite $\{u_n = 2n+1\}$ definie sur $n \in \mathbb{N}$. On veut calculer la somme des u_n :

Démonstration.

$$\begin{aligned} \sum_{k=0}^{n} u_k &= \frac{(u_0 + u_n)(n+1)}{2} \\ &= \frac{(2n+2)(n+1)}{2} \\ &= \frac{2(n+1)(n+1)}{2} \\ &= (n+1)(n+1) \\ &= (n+1)^2 \end{aligned}$$

□

$$\begin{aligned} pH &= -log\left(\frac{[H_3O^+_{(aq)}]}{c_0}\right) \\ [H_3O^+_{(aq)}] &= 10^{-pH} * c_0 \end{aligned}$$

Euler's formula : $e^{\theta i} = \cos(\theta) + i\sin(\theta)$

$$\begin{aligned} x^2 - 2xy = 15 \\ x^2 &= 15 + 2xy \end{aligned}$$

$$\begin{aligned} f(x) &= kx^n \\ f^{(d)}(x) &= \frac{kn!x^{n-d}}{(n-d)!} \\ f(x) &= u(x) * v(x) \\ f^{(d)}(x) &= \sum_{i=0}^{2^{d-1}-1} u^{d-i}v^i + u^i v^{d-i} \end{aligned}$$

$$\begin{aligned} f(x) &= 2x^4 \\ f'(x) &= 8x^3 \\ f''(x) &= 24x^2 \\ f^{(3)}(x) &= 48x \\ f^{(4)}(x) &= 48 \end{aligned}$$